MLP → Girdi $28^2 = 784$
Parameter Count → Layer 1 (Linear: 784 → 128
Weights ⇒ $784 \times 128 = 100,352$
Biases → 128

Layer 2 (Linear 128 → 64)
W ⇒ $128 \times 64 = 8192$ Biases ⇒ 64

Layer 3 (~~Linear~~ Output 64 → 10
W ⇒ $64 \times 10 = 640$ Biases ⇒ 10
Total Parameter: 100,480 + 8256 + 650 = 109,386

Layer 1 ⇒ $2 \times 784 \times 128 \Rightarrow 200,704$ FLOPs
Layer 2 ⇒ $2 \times 128 \times 64 = 16,384$ FLOPs
Layer 3 ⇒ $2 \times 64 \times 10 = 1280$ FLOPs

CNN calculations
Conv2D FLOPs $= 2 \times H_{out} \times W_{out} \times C_{in} \times C_{out} \times K_h \times K_w$

Conv Layer 1 → (32 channels, 3×3 kernel, padding 1)
output size → 28×28
Parameters ⇒ $(1 \times 3 \times 3 \times 32) + 32 = 320$
FLOPs ⇒ $2 \times 28 \times 28 \times 1 \times 32 \times 3 \times 3 = 451,584$ FLOPs

MaxPool Layer (2×2) ⇒ output size becomes 14×14

Conv Layer 2 (32 channels → 64 channels, 3×3 kernels, padding = 1)

Output size ⇒ 14×14

parameters ⇒ (32 × 3 × 3 × 64) + 64 = 18,496

FLOPs ⇒ 2 × 14×14 × 32×64×3×3 = 722,534 FLOPs

Projection 8 Q K V matrices

N = 49 D = 64

Q, K, V Projections ⇒ 3 × (2 × N × D × D)

= 3 × (2 × 49 × 64 × 64) = 307,328 FLOPs

Self-Attention Score Calculation (Q, K^T)

Attention Matrix (N×N) ⇒ 2 × N × N